Fiscal Year:	FY 2014	**Task Last Updated:**	FY 08/21/2014
PI Name:	Donahue, Henry J. Ph.D.		
Project Title:	Integrated Regulation of Bone and Muscle Metabolism by Simulated Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23284-3067	**Congressional District:**	3
Comments:	Formerly at Pennsylvania State University, Hershey, PA, from 2012-2015.		
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	08/01/2012	**End Date:**	07/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lang, Charles (The Pennsylvania State University)		
Grant/Contract No.:	NCC 9-58-MA02802		
Performance Goal No.:			
Performance Goal Text:			

Original project aims/objectives
Our long-term goal is to understand the structural and functional relationship between muscle atrophy (i.e., sarcopenia) and bone loss (i.e., osteopenia) which occurs as a result of extended exposure to the spaceflight environment. The specific objectives of this proposal are to: 1) Examine the time course for structural, functional, and molecular changes occurring in muscle and bone in response to mechanical unloading; 2) Examine the effect of inhibiting, or exaggerating, unloading-induced muscle loss on unloading-induced bone loss; 3) Examine the effect of inhibiting, or exaggerating, unloading-induced bone loss on unloading-induced muscle loss and 4) Evaluate potential therapeutic strategies for prevention of muscle and bone loss during unloading. Upon completion of our specific aims, and based on our experimental approaches, we will recommend specific therapeutic strategies and countermeasures that target muscle and bone as an integrated unit and respect their inherent interdependency. To accomplish these objectives, we will use hindlimb suspension (HLS), a well established in vivo model of simulated microgravity unloading. Our overall

Task Description:	hypothesis is that unloading-induced osteopenia is at least partially dependent on unloading-induced sarcopenia and, conversely, unloading induced sarcopenia is at least partially dependent on unloading-induced osteopenia. The rationale for this proposed research is that there has not been a comprehensive and simultaneous examination of the effects of unloading on both muscle and bone. Understanding how these two systems respond to simulated microgravity will allow for the development of effective countermeasures to preserve astronaut health during missions and upon return to Earth. The research team assembled has extensive experience examining, both in vitro and in vivo, bone (Donahue) and muscle (Lang) response to loading and unloading. This collective expertise will bring a unique and valuable perspective to the project. Key findings The second year of the project has proceeded very well with no unanticipated problems. Results of Aim 1 have been published in the Journal of Bone and Mineral Research. The manuscript is now undergoing revision. We have also completed aim 3 and are currently analyzing results. In completing aim 3 4-month-old wild type C57BL/6J mice or C57BL/6J mice globally deficient in Sost since birth (Sost KO), C57BL/6J mice were subjected to HLS or maintained in control cages for 2 weeks (4 groups, n=4-9/group). Mice were subjected to baseline and endpoint μCT scans. Total body weight and quadriceps weight were recorded and per cent muscle and fat quantified by 1H-NMR. At baseline, Sost KO mice displayed significantly greater trabecular bone volume relative to WT. Loss of trabecular bone following 2 weeks of unloading was significantly attenuated in Sost KO. Sost KO displayed reduced loss of trabecular BV/TV (-4% vs. -47%, $p<0.05$), trabecular thickness (-2% vs. -41%, $p<0.05$), and trabecular number (+6% vs. -15%, $p<0.05$). All mice were of similar weight at the start of study; however, Sost KO displayed significantly more adipose tissue and less lean tissue than WT mice. HLS resulted in a further gain of fat and loss of muscle in Sost KO. HLS resulted in decreased quadriceps weight in WT. Quadricep weight was decreased in Sost KO controls relative to WT controls but did not decrease further after HLS. These are the first studies to examine skeletal muscle in Sost KO mice exposed to unloading. While Sost KO gained bone and were resistant to unloading, they displayed dramatic sarcopenia and increased adiposity. Impact Previous studies did not observe sarcopenia in mice given anti sclerostin antibody for three weeks nor were these mice resistant to HLS-induced muscle loss. Our results suggest that long term sclerostin deficiency, while having positive effects on bone, may have deleterious effects on muscle. If confirmed, these findings should be considered when developing therapeutic protocols using sclerostin antibody. Proposed research plan for the coming year. During the coming year we will complete aim 3 and begin aim 2 and 4.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	On the ground, concomitant bone and muscle loss are frequently experienced by individuals confined to bed for long durations, whether a result of neurological injury, trauma, or neuromuscular disease (i.e., cerebral palsy, muscular dystrophy, etc.). In these patients, and in healthy patients subjected to extended bed rest, a significant loss of muscle and bone occurs. In the case of recoverable disease, or when ambulation is still possible, this leaves patients at risk for falls and catastrophic fracture. Furthermore, age-related bone and muscle loss is a serious problem in our increasingly aged population. Understanding the interdependence of muscle and bone loss during spaceflight, and the development of countermeasure strategies, will provide a framework for the development of preventative treatments for astronauts and the clinical population, especially those suffering from osteopenia and sarcopenia.
Task Progress:	We have completed aims 1 and 3 and have begun aim 2. Results from aim 1 have been published and results of aim 3 are being prepared for publication.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Abstracts for Journals and Proceedings	Lloyd SA, Loiselle AE, Zhang Y, Donahue HJ. "Connexin 43 deficiency protects against unloading-induced osteocytic osteolysis." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3019.pdf , Feb-2014
Abstracts for Journals and Proceedings	Lloyd SA, Zhang Y, Paul EM, Laufenberg LJ, Lang CH, Donahue HJ. "Interdependence of Muscle and Bone Loss Induced by Disuse and Aging." 35th Annual Meeting of the American Society for Bone and Mineral Research, Baltimore, Maryland, October 4-7, 2013. J Bone Miner Res. 2013;28 (Suppl 1). , Oct-2013
Articles in Peer-reviewed Journals	Lloyd SA, Loiselle AE, Zhang Y, Donahue HJ. "Evidence for the role of connexin 43-mediated intercellular communication in the process of intracortical bone resorption via osteocytic osteolysis." BMC Musculoskelet Disord. 2014 Apr 9;15:122. http://dx.doi.org/10.1186/1471-2474-15-122 ; PubMed PMID: 24716486; PubMed Central PMCID: PMC3984635 , Apr-2014
Articles in Peer-reviewed Journals	Lloyd SA, Loiselle AE, Zhang Y, Donahue HJ. "Shifting paradigms on the role of connexin43 in the skeletal response to mechanical load." Journal of Bone and Mineral Research. 2014 Feb;29(2):275-86. PubMed PMID: 24588015 , Feb-2014
Articles in Peer-reviewed Journals	Lloyd SA, Lang CH, Zhang Y, Paul EM, Laufenberg LJ, Lewis GS, Donahue HJ. "Interdependence of muscle atrophy and bone loss induced by mechanical unloading." Journal of Bone and Mineral Research. 2014 May;29(5):1118-30. http://dx.doi.org/10.1002/jbmr.2113 ; PubMed PMID: 24127218; PubMed Central PMCID: PMC4074925 , May-2014